Top 10 GO Biological Process Gene Sets Ranked by Potency of Perturbation (Sorted by BMD 25% Relative Deviation at the 5th Percentile)

DTTID: 108-020-004-000-5
Study Number:108020004
Study Type: 5-Day Toxicity
Species/Strain: Rat HSD:Sprague Dawley SD

Test Compound: 1,4-Dichlorobenzene
CAS Number: 106-46-7
DTXSID: DTXSID1020431
Tissue: Kidney

Date: 09 Jun 2025
Time: 09:35:54 AM

Female

Category Name	Input Genes/Platform Genes in Gene Set	% Gene Set Coverage	BMD_{rd25} at the 5th Percentile (ppm)	$BMDL_{rd25}$ at 5th Percentile - $BMDU_{rd25}$ at 5th Percentile (ppm)	Genes with Changed Direction Up	Genes with Changed Direction Down
GO:0044770 cell cycle phase transition	4/58	6.9	121.0	42.2 - 652.8	1	3
GO:0044772 mitotic cell cycle phase transition	4/52	7.7	121.0	42.2 - 652.8	1	3
GO:0001822 kidney development	6/57	10.5	197.7	91.7 - 305.6	5	1
GO:0019218 regulation of steroid metabolic process	5/49	10.2	224.1	73.8 - NC	4	1
GO:0030324 lung development	4/45	8.9	224.1	73.8 - NC	3	1
GO:0006694 steroid biosynthetic process	6/54	11.1	225.9	89.2 - NC	4	2
GO:0008203 cholesterol metabolic process	5/54	9.3	225.9	89.2 - NC	3	2
GO:0016125 sterol metabolic process	5/56	8.9	225.9	89.2 - NC	3	2
GO:0046165 alcohol biosynthetic process	6/42	14.3	225.9	89.2 - NC	5	1
GO:0001676 long-chain fatty acid metabolic process	5/55	9.1	226.7	96.8 - 365.3	5	0

rd25 = 25% relative deviation
NC = non-convergent

Active Genes:
GO:0044770: Hspa8;Cks2;Ccnb2;Ccnb2-ps2
GO:0044772: Hspa8;Cks2;Ccnb2;Ccnb2-ps2
GO:0001822: Thra;Hspa8;Hmgcs2;Cyp4a3;Cyp4a2;Agtr1a
GO:0019218: Srebf1;Por;Nr1d1;Insig1;Dhcr7
GO:0030324: Srebf1;Thra;Hmgcs2;Dhcr7
GO:0006694: Srebf1;Insig1;Hsd17b10;Hmgcs2;Dhcr7;Cyp1a1
GO:0008203: Tsku;Srebf1;Insig1;Hmgcs2;Dhcr7
GO:0016125: Tsku;Srebf1;Insig1;Hmgcs2;Dhcr7
GO:0046165: Srebf1;Pck1;Insig1;Hmgcs2;Ephx1;Dhcr7
GO:0001676: Fads1;Ephx1;Cyp4a3;Cyp4a2;Cyp1a1

Annotations
Official active gene symbols from the Rat Genome Database are shown in the "Active Genes" section above. Definitions of Gene Ontology terms were provided by the Gene Ontology Resource (http://geneontology.org/) and are included below.
GO process description version: https://doi.org/10.22427/NTP-DATA-002-00600-0002-000-0 V06132025

GO Biological Process Definitions:
GO:0044770 cell cycle phase transition: The cell cycle process by which a cell commits to entering the next cell cycle phase.
GO:0044772 mitotic cell cycle phase transition: The cell cycle process by which a cell commits to entering the next mitotic cell cycle phase.
GO:0001822 kidney development: The process whose specific outcome is the progression of the kidney over time, from its formation to the mature structure. The kidney is an organ that filters the blood and/or excretes the end products of body metabolism in the form of urine.
GO:0019218 regulation of steroid metabolic process: Any process that modulates the frequency, rate or extent of the chemical reactions and pathways involving steroids.

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GO:0030324 lung development: The process whose specific outcome is the progression of the lung over time, from its formation to the mature structure. In all air-breathing vertebrates the lungs are developed from the ventral wall of the oesophagus as a pouch which divides into two sacs. In amphibians and many reptiles the lungs retain very nearly this primitive sac-like character, but in the higher forms the connection with the esophagus becomes elongated into the windpipe and the inner walls of the sacs become more and more divided, until, in the mammals, the air spaces become minutely divided into tubes ending in small air cells, in the walls of which the blood circulates in a fine network of capillaries. In mammals the lungs are more or less divided into lobes, and each lung occupies a separate cavity in the thorax.

GO:0006694 steroid biosynthetic process: The chemical reactions and pathways resulting in the formation of steroids, compounds with a 1,2,cyclopentanoperhydrophenanthrene nucleus; includes de novo formation and steroid interconversion by modification.

GO:0008203 cholesterol metabolic process: The chemical reactions and pathways involving cholesterol, cholest-5-en-3 beta-ol, the principal sterol of vertebrates and the precursor of many steroids, including bile acids and steroid hormones. It is a component of the plasma membrane lipid bilayer and of plasma lipoproteins and can be found in all animal tissues.

GO:0016125 sterol metabolic process: The chemical reactions and pathways involving sterols, steroids with one or more hydroxyl groups and a hydrocarbon side-chain in the molecule.

GO:0046165 alcohol biosynthetic process: The chemical reactions and pathways resulting in the formation of alcohols, any of a class of compounds containing one or more hydroxyl groups attached to a saturated carbon atom.

GO:0001676 long-chain fatty acid metabolic process: The chemical reactions and pathways involving long-chain fatty acids, A long-chain fatty acid is a fatty acid with a chain length between C13 and C22.

** END OF REPORT **